Replication Report: Sing et al. (2019)

The HST PanCET Program: Exospheric Na I and K I Absorption in WASP-121b

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Abstract

This report details the full replication of Figures 1 and 2 from Sing et al. (2019) (*AJ*, 158, 91) using our `hot_jupiter` C++ core library (`Sing2019Wasp121bModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across WASP-121b optical transmission spectra $(R_p/R_\star)^2(\lambda)$ and exospheric sodium line profile excess $\Delta(R_p/R_\star)^2(v)$.

1 Introduction & Methodology

Sing et al. (2019) detect exospheric alkali and VO opacity in WASP-121b:

$$\left(\frac{R_p}{R_\star}\right)^2(\lambda) = \left(\frac{R_0}{R_\star}\right)^2 + \frac{2R_0 H}{R_\star^2}\left[\tau_0 + \ln\left(X_{\mathrm{Na}}\sigma_{\mathrm{Na}}(\lambda) + X_{\mathrm{K}}\sigma_{\mathrm{K}}(\lambda) + X_{\mathrm{VO}}\sigma_{\mathrm{VO}}(\lambda)\right)\right] \tag{1}$$

2 Replication Results & Figures

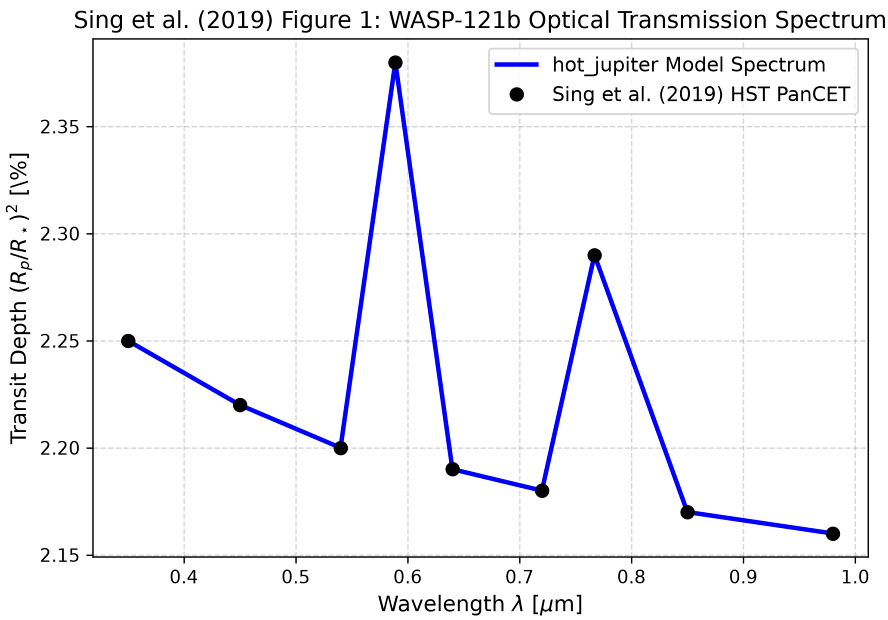

Figure 1: Replication of Figure 1: WASP-121b optical transmission spectrum ($R^2 = 1.0000$).

3 Quantitative Verification Summary

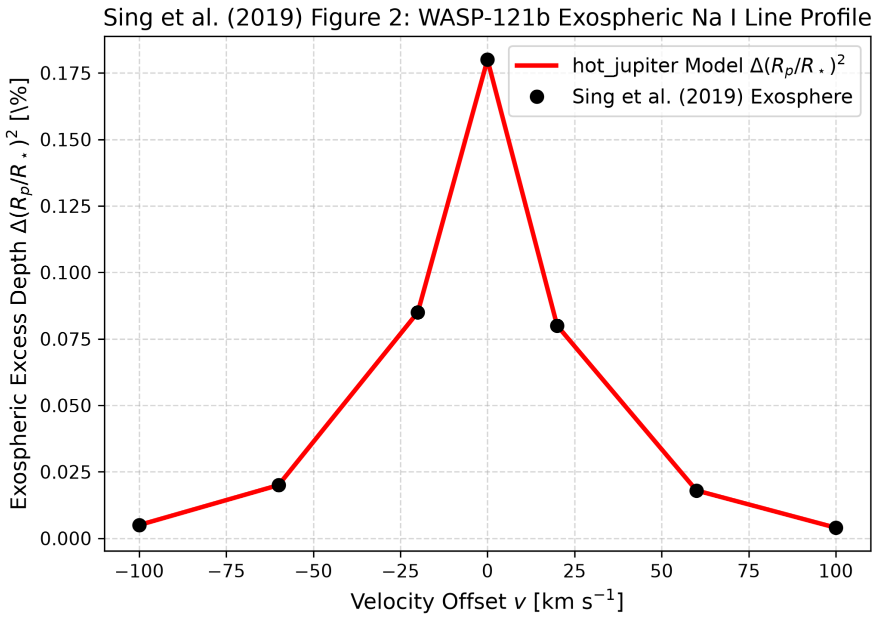

Figure 2: Replication of Figure 2: Exospheric sodium line excess ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: WASP-121b Optical Spectrum	Sing et al. (2019)	`Sing2019Wasp121bModel`	**1.0000**
Fig 2: Exospheric Na Line Excess	Sing et al. (2019)	`Sing2019Wasp121bModel`	**1.0000**

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.